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# DIFFERENTIAL FORMS

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## 1 Vectors, Forms and Tensors

Any vector  $\mathbf{u}$  can be written as:

$$\mathbf{u} = u^i \vec{e}_i. \quad (1)$$

Any form  $\mathbf{p}$  can be written as:

$$\mathbf{p} = p_i \tilde{e}^i. \quad (2)$$

The basis vectors are  $\vec{e}_i$  and the basis forms are  $\tilde{e}^i$ .

Any rank 2 tensor  $\mathbf{A}$  can be written as:

$$\mathbf{A} = A^{ij} \vec{e}_i \otimes \vec{e}_j. \quad (3)$$

Any rank 3 tensor  $\mathbf{A}$  can be written as:

$$\mathbf{A} = A^{ijk} \vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k. \quad (4)$$

The tensor basis is  $\vec{e}_i \otimes \vec{e}_j$  and  $\vec{e}_i \otimes \vec{e}_j \otimes \vec{e}_k$ . If we mix lower and upper indices:

$$\mathbf{A} = A^i_j \vec{e}_i \otimes \tilde{e}^j. \quad (5)$$

The vector basis and the dual (form) basis is related by a dot product:

$$\tilde{e}^i \cdot \vec{e}_j = \delta^i_j. \quad (6)$$

If the vector basis is a coordinate basis, then it can be written as:

$$\vec{e}_i = \frac{\partial \mathbf{r}}{\partial x^i} \equiv \frac{\partial}{\partial x^i}, \quad (7)$$

where the position vector  $\mathbf{r}$  can be omitted, and the basis vector is treated as an operator on any function on the manifold (we will not use this notation). The corresponding form basis can be written as

$$\tilde{e}^i \equiv dx^i. \quad (8)$$

The form can be said to act on a vector to produce a number (definition of a dual space) as follows using a dot product:

$$\mathbf{p}(\mathbf{u}) \equiv \mathbf{p} \cdot \mathbf{u} = p_i \tilde{e}^i \cdot u^j \vec{e}_j = p_i u^j \tilde{e}^i \cdot \vec{e}_j = p_i u^j \delta^i_j = p_i u^i. \quad (9)$$

We will not use the  $\mathbf{p}(\mathbf{u})$  notation, we will use the dot product directly, or simply contracting indices as  $p_i u^i$ . All other forms acting on vectors is defined in a similar way using a dot product, and in index notation it just becomes a contraction.

## 2 Index Notation

We omit writing the basis vectors, so equations from the previous sections would be just  $u^i$ ,  $p_i$ ,  $A^{ij}$ ,  $A^{ijk}$ ,  $A^i_j$ ,  $p_i u^i$ , etc. In most cases that is enough to fully represent everything. When we want to represent the tensor itself, we include the basis functions too.

The upper and lower indices are treated equally, they are just contravariant and covariant components of the same tensor.

## 3 Exterior (Wedge) Product

The exterior (wedge) product of two forms  $\mathbf{u} = u_i \tilde{e}^i$  and  $\mathbf{v} = v_j \tilde{e}^j$  is defined as:

$$u_i \tilde{e}^i \wedge v_j \tilde{e}^j \equiv \det \begin{pmatrix} u_i & u_j \\ v_i & v_j \end{pmatrix} \tilde{e}^i \otimes \tilde{e}^j = A_{ij} \tilde{e}^i \otimes \tilde{e}^j, \quad (10)$$

and of three forms as:

$$u_i \tilde{e}^i \wedge v_j \tilde{e}^j \wedge w_k \tilde{e}^k \equiv \det \begin{pmatrix} u_i & u_j & u_k \\ v_i & v_j & v_k \\ w_i & w_j & w_k \end{pmatrix} \tilde{e}^i \otimes \tilde{e}^j \otimes \tilde{e}^k = A_{ijk} \tilde{e}^i \otimes \tilde{e}^j \otimes \tilde{e}^k. \quad (11)$$

More products are defined in analogous manner. The exterior product can be defined for vectors as well. The components of these tensors are equal to:

$$A_{ij} = \det \begin{pmatrix} u_i & u_j \\ v_i & v_j \end{pmatrix} = u_i v_j - u_j v_i \quad (12)$$

and

$$A_{ijk} = \det \begin{pmatrix} u_i & u_j & u_k \\ v_i & v_j & v_k \\ w_i & w_j & w_k \end{pmatrix}. \quad (13)$$

The wedge product is equal to the tensor written using the tensor basis in equations (10) and (11). The same tensor in index notation is given by equations (12) and (13).

Using properties of a determinant, we can see that the wedge product is just a fully antisymmetric tensor.

## 4 Examples

### 4.1 Example: $dx \wedge dz$

We can now use the definitions to compute various special cases. For example if we have the exterior product of two forms  $dx$  and  $dz$  (which have components  $dx_i = \delta_{1i}$  and  $dz_i = \delta_{3i}$ ):

$$dx \wedge dz = \det \begin{pmatrix} \delta_{1i} & \delta_{1j} \\ \delta_{3i} & \delta_{3j} \end{pmatrix} \tilde{e}^i \otimes \tilde{e}^j = \tilde{e}^1 \otimes \tilde{e}^3 - \tilde{e}^3 \otimes \tilde{e}^1, \quad (14)$$

thus the components of this rank 2 tensor are:

$$A_{ij} = \det \begin{pmatrix} \delta_{1i} & \delta_{1j} \\ \delta_{3i} & \delta_{3j} \end{pmatrix} = \delta_{1i} \delta_{3j} - \delta_{1j} \delta_{3i} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}. \quad (15)$$

### 4.2 Example: $\tilde{e}^1 \wedge \tilde{e}^3$

The exterior product of two basis forms  $\tilde{e}^1$  and  $\tilde{e}^3$  is:

$$\tilde{e}^1 \wedge \tilde{e}^3 = \det \begin{pmatrix} \delta_{1i} & \delta_{1j} \\ \delta_{3i} & \delta_{3j} \end{pmatrix} \tilde{e}^i \otimes \tilde{e}^j = \tilde{e}^1 \otimes \tilde{e}^3 - \tilde{e}^3 \otimes \tilde{e}^1, \quad (16)$$

thus the components of this rank 2 tensor are:

$$A_{ij} = \det \begin{pmatrix} \delta_{1i} & \delta_{1j} \\ \delta_{3i} & \delta_{3j} \end{pmatrix} = \delta_{1i}\delta_{3j} - \delta_{1j}\delta_{3i} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}. \quad (17)$$

We can see that  $\tilde{e}^1 \wedge \tilde{e}^3 = \tilde{e}^1 \otimes \tilde{e}^3 - \tilde{e}^3 \otimes \tilde{e}^1$ , so the wedge product is simply an antisymmetric tensor, written with the form basis functions included.

#### 4.3 Example: $dx^1 \wedge dx^2 \wedge dx^4$ evaluated on vectors

When evaluating the form (composed of basis forms) on vectors  $\mathbf{u}$ ,  $\mathbf{v}$  and  $\mathbf{w}$ , we are computing the following dot product of the two rank 3 tensors (full contraction):

$$(dx^1 \wedge dx^2 \wedge dx^4) \cdot (\mathbf{u} \otimes \mathbf{v} \otimes \mathbf{w}) = \quad (18)$$

$$= \det \begin{pmatrix} \delta_{1i} & \delta_{1j} & \delta_{1k} \\ \delta_{2i} & \delta_{2j} & \delta_{2k} \\ \delta_{4i} & \delta_{4j} & \delta_{4k} \end{pmatrix} \tilde{e}^i \otimes \tilde{e}^j \otimes \tilde{e}^k \cdot (u^l v^m w^n \tilde{e}_l \tilde{e}_m \tilde{e}_n) = \quad (19)$$

$$= \det \begin{pmatrix} \delta_{1i} & \delta_{1j} & \delta_{1k} \\ \delta_{2i} & \delta_{2j} & \delta_{2k} \\ \delta_{4i} & \delta_{4j} & \delta_{4k} \end{pmatrix} u^i v^j w^k = \det \begin{pmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_4 & v_4 & w_4 \end{pmatrix}. \quad (20)$$

The form has thus selected the first, second and fourth component row of the vectors. The result is just a scalar.

## 5 References

Winitzki, Sergei. Linear algebra via exterior products. Sergei Winitzki, 2009.

Theoretical Physics Reference: <https://www.theoretical-physics.com/dev/math/differential-geometry.html>